



Specifying and Analyzing Transactional Consistency Models with Predicates

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A semantic and graph-theoretic foundation for predicate operations

Item operations vs. predicate operations

Read(x, v)

Read(2, 60)

Write(x, v)

Write(3, 40)

point access
one value

Employee

EID	Salary
0	80
1	100
2	60
3	40

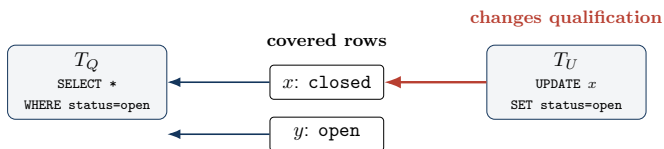
PredRead(P, M)

$P : \text{Salary} < 80$

condition-based access
a set of values

One query can observe both matching and non-matching employees.

A predicate result contains more than returned rows



result: $\{y\}$

Key observation: $x \notin \text{result}$
still records an observation of x .

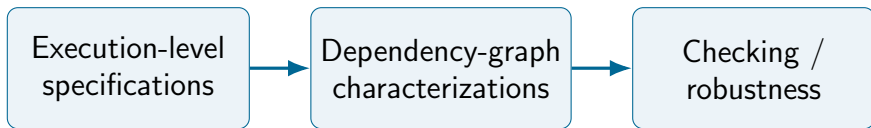
Item reads: value observed

Predicate reads: returned rows and excluded rows

Phantom-style interactions need an explanation for both.

Berenson et al., 1995; Adya, 1999

The gap: familiar foundations stop at named items



Predicate operations are missing from the semantic-graph correspondence.

Question: Can we recover an exact, predicate-aware foundation for SER and SI?

Semantic anchor: explain every covered item

VIS/AR execution framework

Visible transactions

provide the version set



Latest visible writer

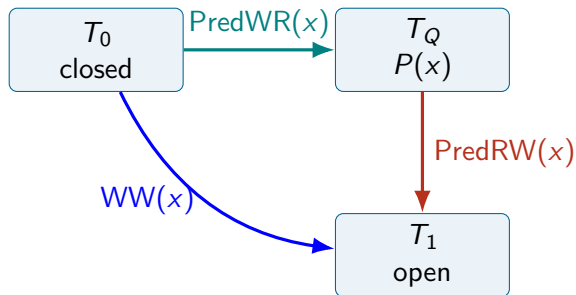
explains each covered item

Returned item: witness matches P

Omitted item: witness does not match P

Cerone, Bernardi, and Gotsman, 2015

From a semantic witness to graph edges



PredWR: version witness

PredRW: later writer changes the observed result

Result-sensitive: $\Delta(T_0, T_Q, T_1, x)$

The graph records why the result was observed — including the absence of x .

Exact graph characterizations

Serializability

$$\mathcal{H} \models \text{SER}$$



internally consistent +
acyclic predicate-aware graph

Snapshot isolation

$$\mathcal{H} \models \text{SI}$$



internally consistent +
every cycle has **two adjacent**
RW / PredRW edges

Not analogy: both equivalences are proved against the predicate-aware axioms.

Cerone and Gotsman, 2018

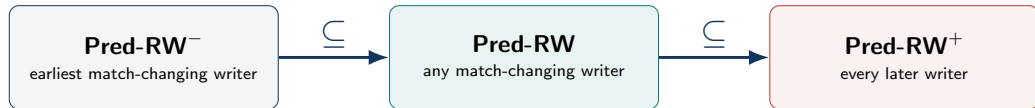
Why the proof bridge matters



observable histories $\longleftrightarrow$ executions

- Predicate graphs are **observable-history** objects.
- Axiomatics supplies the **semantic** correctness target.
- The bridge validates graph-based analysis beyond item-only workloads.

Related work disagrees on one crucial edge



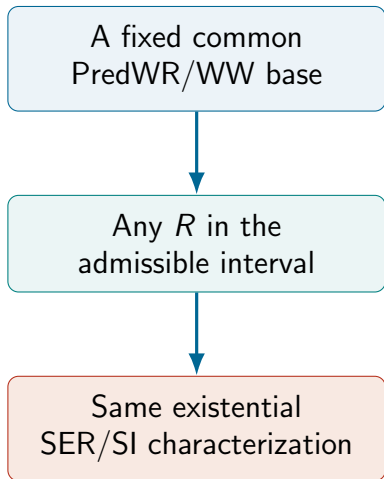
Same existential history-level characterization of SER and SI
edge sets and checking cost may differ

Sun and Zou, 2025

Adya, 1999; "Generalized Isolation
Level Definitions" 2000

Bouajjani, Enea, and Román-Calvo,
2025

What the design space tells us



The theorem is semantic, not a claim that edge sets coincide.

Practical consequence: choose an admissible edge policy for an analysis, while retaining its history-level meaning.

Checking cost can still differ.

Future work: turn the foundation into analysis tools

Efficient black-box
checking

Recover version sets
and version order

Robustness analysis

Extend static reasoning
to predicate operations

Both need a semantics-preserving predicate dependency graph.

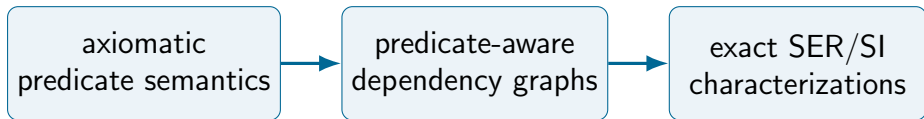
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Predicate results are observations over a covered set.



A principled basis for checking and analyzing databases with predicates.